Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: To apply a suitable sequence of morphological operations**	
Open-ended-Problem: 06	**Date:02-11-2026**	**Enrollment No:92400133121**

Aim:

To apply a suitable sequence of morphological operations with properly chosen structuring elements on three images, so that small bright noise is removed, narrow dark cracks and holes are filled, and the overall shape of the object is preserved.

Theory:

Dilation, erosion, opening, and closing are morphological operations used in image processing to manipulate the shape and size of objects in an image.

Dilation:
Dilation expands the boundaries of objects in an image by adding pixels to the object boundaries. It is achieved by sliding a structuring element over the image and assigning the maximum pixel value within the neighborhood of each pixel.

Erosion:
Erosion shrinks the boundaries of objects in an image by removing pixels from the object boundaries. It is achieved by sliding a structuring element over the image and assigning the minimum pixel value within the neighborhood of each pixel.

Opening:
Opening is an erosion operation followed by a dilation operation. It is used to remove noise and small objects from the image while preserving the larger objects shapes.

Closing:
Closing is a dilation operation followed by an erosion operation. It is used to close small gaps and holes within objects while preserving the overall object shapes.

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

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```
# Read Image 1
image1 = cv2.imread("images/Image_1.png", 0)

# Read Image 2
image2 = cv2.imread("images/Image_2.png", 0)

# Read Image 3
image3 = cv2.imread("images/Image_3.png", 0)

plt.figure(figsize=(12, 5))

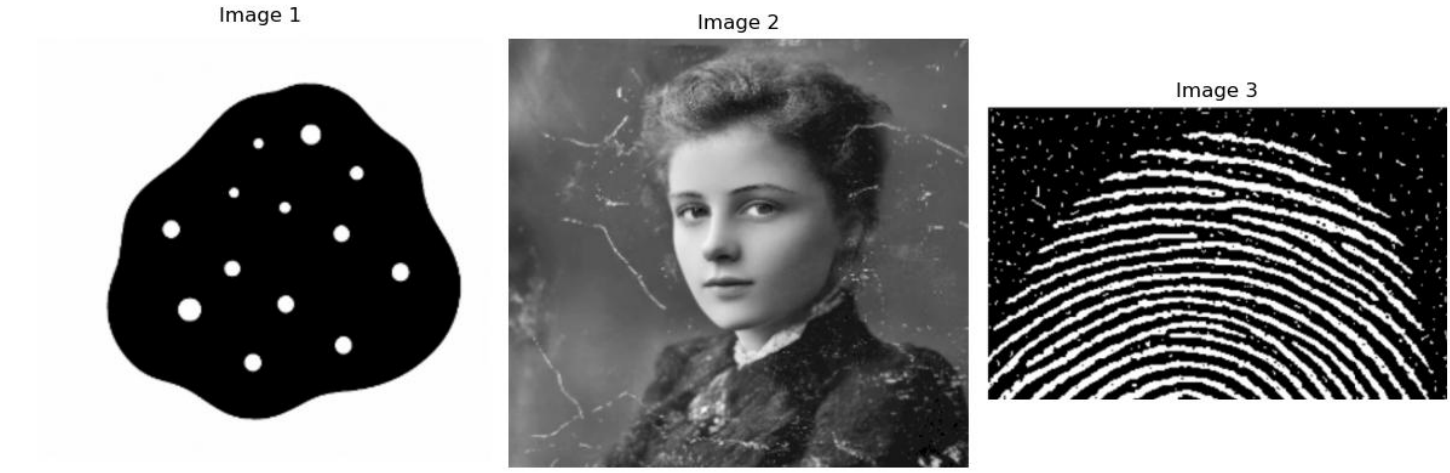
plt.subplot(1, 3, 1)
plt.imshow(image1, cmap="gray")
plt.title("Image 1")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(image2, cmap="gray")
plt.title("Image 2")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(image3, cmap="gray")
plt.title("Image 3")
plt.axis("off")

plt.tight_layout()
plt.show()
```

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```
# Kernel for Image 1
kernel1 = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (35, 35))

# Binary conversion
_, binary1 = cv2.threshold(image1, 127, 255,cv2.THRESH_BINARY)

# Invert image
inverted1 = cv2.bitwise_not(binary1)

# Apply closing
closed1 = cv2.morphologyEx(inverted1,cv2.MORPH_CLOSE,kernel1)

# Invert back
result1 = cv2.bitwise_not(closed1)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.title("Original Image 1")
plt.imshow(image1, cmap="gray")
plt.axis("off")

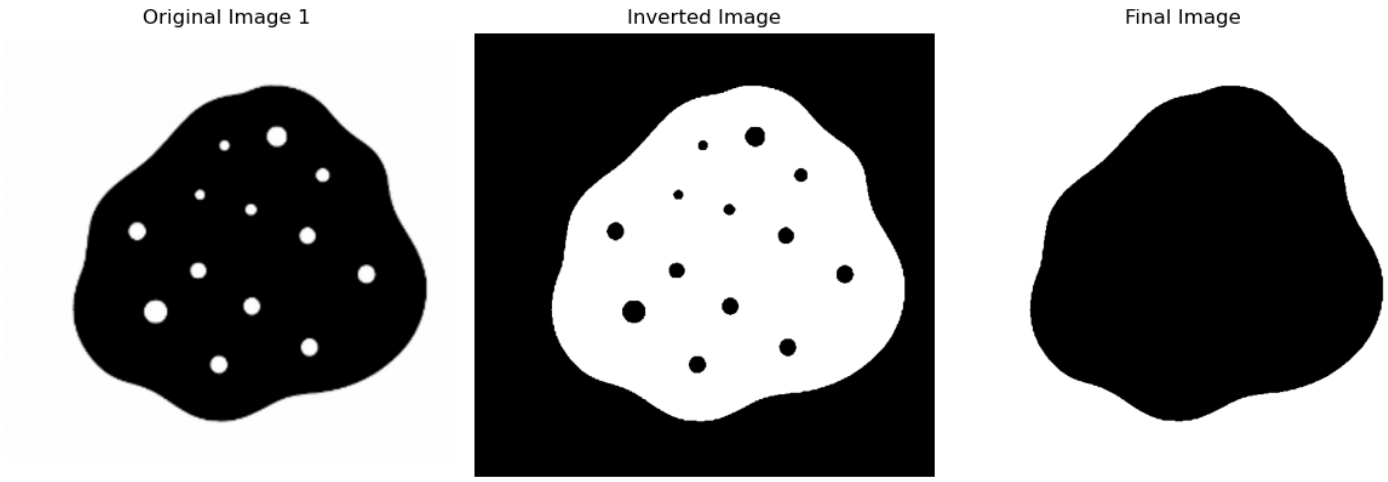
plt.subplot(1, 3, 2)
```

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```
plt.title("Inverted Image")
plt.imshow(inverted1, cmap="gray")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.title("Final Image")
plt.imshow(result1, cmap="gray")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Observation:

- ❖ Binary Image with a black object containing white circular holes.
- ❖ After inversion the holes become black gaps inside a white object.
- ❖ The 35x35 ellipse filled all of them and the outline stayed the same.

Conclusion:

- ✓ Closing fills holes inside an object.

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- ✓ The kernel must be bigger than the biggest hole, and the ellipse keeps the smooth shape.

```
# Kernel for Image 2
kernel2_open = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (9, 3))

kernel2_close = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))

# Apply opening to remove small bright noise
opened2 = cv2.morphologyEx(image2,cv2.MORPH_OPEN,kernel2_open)

# Apply closing to fill narrow dark cracks
result2 = cv2.morphologyEx(opened2,cv2.MORPH_CLOSE,kernel2_close)

plt.figure(figsize=(12, 4))

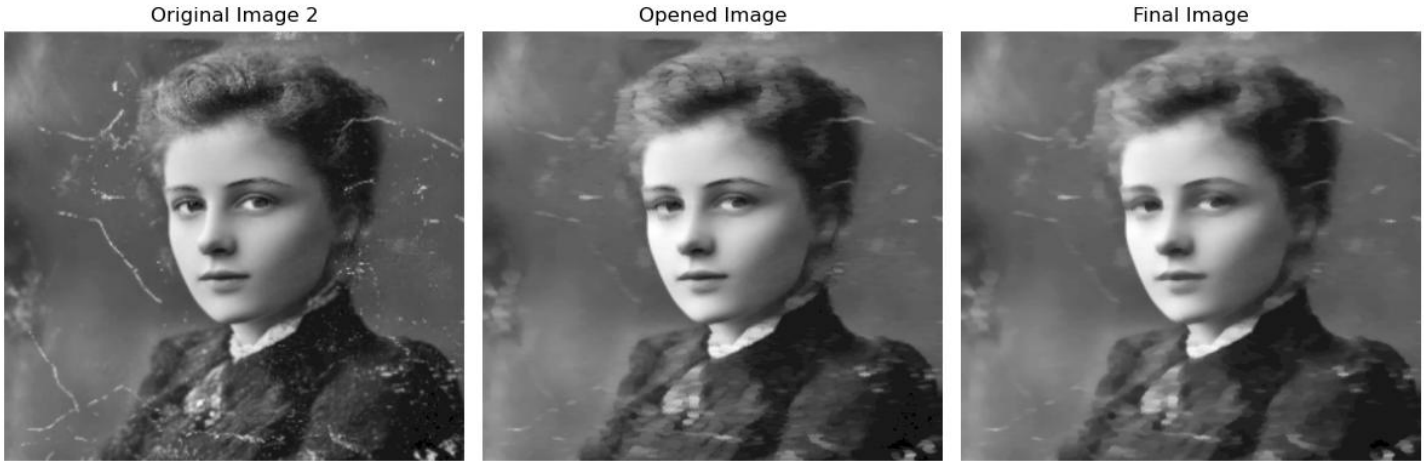
plt.subplot(1, 3, 1)
plt.title("Original Image 2")
plt.imshow(image2, cmap="gray")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.title("Opened Image")
plt.imshow(opened2, cmap="gray")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.title("Final Image")
plt.imshow(result2, cmap="gray")
plt.axis("off")

plt.tight_layout()
plt.show()
```

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Observation:

- ❖ Grayscale photo so no thresholding was applied.
- ❖ It has bright scratched and specks and some dark circles.
- ❖ Opening with 9x9 removed them and closing with 5x5 filled the dark gaps.
- ❖ The face is clear but the image is slightly blurred.

Conclusion:

- ✓ Opening removes the bright noise first, then closing fills the dark cracks.
- ✓ Morphology blurs fine detail, so it cannot fully restore a real photo.

```
# Kernel for Image 3
kernel3 = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))

# Binary conversion
_, binary3 = cv2.threshold(image3, 127, 255,cv2.THRESH_BINARY)

# Remove small bright noise
opened3 = cv2.morphologyEx(binary3,cv2.MORPH_OPEN,kernel3)
```

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```
# Fill narrow dark cracks
result3 = cv2.morphologyEx(opened3,cv2.MORPH_CLOSE,kernel3)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.title("Original Image 3")
plt.imshow(image3, cmap="gray")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.title("Opened Image")
plt.imshow(opened3, cmap="gray")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.title("Final Image")
plt.imshow(result3, cmap="gray")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Observation:

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- ❖ Binary image with white ridges on a black background, with white specks and small ridge breaks.
- ❖ Opening removed the specks. A 5×5 kernel cut thin ridges into pieces, while 3×3 kept them connected.
- ❖ Closing repaired the small breaks.

Conclusion:

- ✓ Opening then closing cleans the noise and repairs the ridges.
- ✓ Small kernels are needed because the ridge gaps are very narrow.

Overall Conclusion:

- ➢ Opening removes small bright noise.
- ➢ Closing fills small dark cracks and holes.
- ➢ Opening is done first so closing does not join the noise to the object.
- ➢ The kernel must be bigger than the defect but smaller than the details to keep.
- ➢ The best combination depends on the type and size of the defect in each image.